Sangam Parajuli

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EDUCATION

Thapathali Campus, Institute of Engineering, Tribhuvan University May 2022 – July 2026
Bachelors in Computer Engineering *Kathmandu, Nepal*

EXPERIENCE

Wiseyak Inc.
AI Research Intern — Multilingual Speech Processing *December 2025 – March 2026*

- Contributing to the Speech team in building multilingual end-to-end Automatic Speech Recognition (ASR) and Text-to-Speech (TTS) systems with support for English, Nepali, Hindi, and Maithili .
- Led data refinement pipelines to ensure high-quality, clean training corpora for multilingual ASR model development.
- Performed domain classification to organize and categorize speech data across diverse topics, improving model generalization and robustness.
- Integrated the AI model with FastAPI to develop fully working app also used Docker for continous development

PUBLICATIONS/PREPRINTS

- Racing Versus Routing for Hiding DRAM Refresh Stalls with Cross-Channel Replicas
  DOI: 10.2139/ssrn.7537683
- Efficiency Over Scale: A Lightweight Approach to Nepali Automatic Speech Recognition
  DOI: 10.13140/RG.2.2.33145.63849 | Paper | Model | Dataset | Noisy Benchmark

PROJECTS

Nepali and English Multilingual Automatic Speech Recognition GitHub
English, Nepali, Conformer, GPT-2, FAISS, ONNX *Hugging Face* | *Paper*

- Designed a modular English-Nepali conversational AI pipeline with swappable ASR, language modeling, RAG, translation, and speech synthesis components.
- Fine-tuned Conformer ASR models, achieving 6.14% WER on LibriSpeech and 31.40% WER / 11.51% CER on Nepali FLEURS, surpassing Whisper variants with fewer parameters.
- Built a GPT-2 language model with FAISS-backed RAG on Nepali news, reaching 0.9303 BERTScore-F1 and 0.5862 ROUGE-L.

LLM from scratch GitHub
Python & PyTorch

- Developed a transformer-based language model from scratch using PyTorch including core components like multi-head self-attention, layer normalization, and feed-forward block.
- Reproduced the GPT-2 architecture with adaptable hyperparameters to load pretrained weights.
- Implemented and integrated top-k sampling and temperature sampling during inference.
- Made a configurable architecture with custom training support via autoregressive loss, gradient clipping, etc.

Graphical Content Recognition with Editable Charts and Layout Retention GitHub
Python, Detectron2, PaddleOCR, OpenCV, PyTorch *Jupyter Notebooks*

- Built a machine learning model to detect, extract, and reconstruct graphical elements like charts and tables from document images, preserving the original layout.
- Utilized Python with Detectron2 for object detection, PaddleOCR for text recognition, and OpenCV for image processing.
- Developed custom algorithms to accurately extract and recreate editable graphical content, improving precision over existing methods.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, C#
- **Frameworks & Libraries:** Qt, Django, PyTorch, Flask, PaddleOCR, Detectron2, OpenCV, NumPy, pandas, TensorFlow
- **Tools & Platforms:** Git, GitHub, Render, Supabase
- **Other:** Game Development (Unity), Data Handling, HTML/CSS/JavaScript